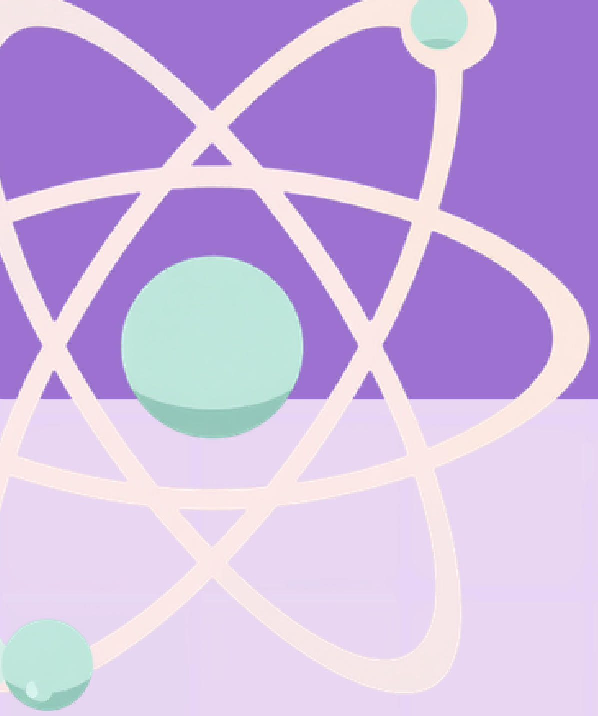
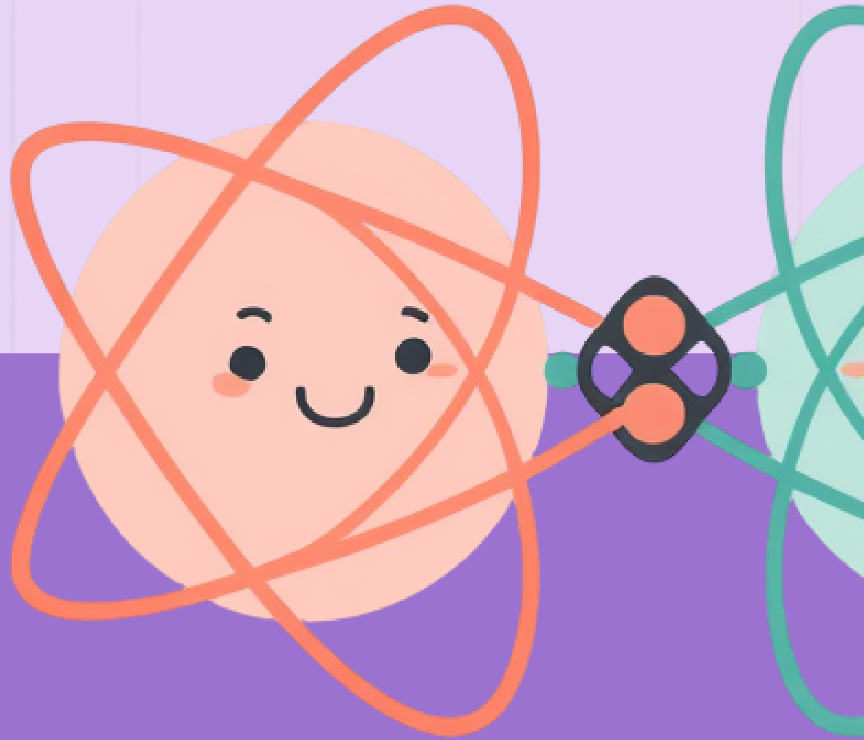


CHEMICAL BONDING: IONIC VS COVALENT

A Fun Exploration for Grade 6 STEM

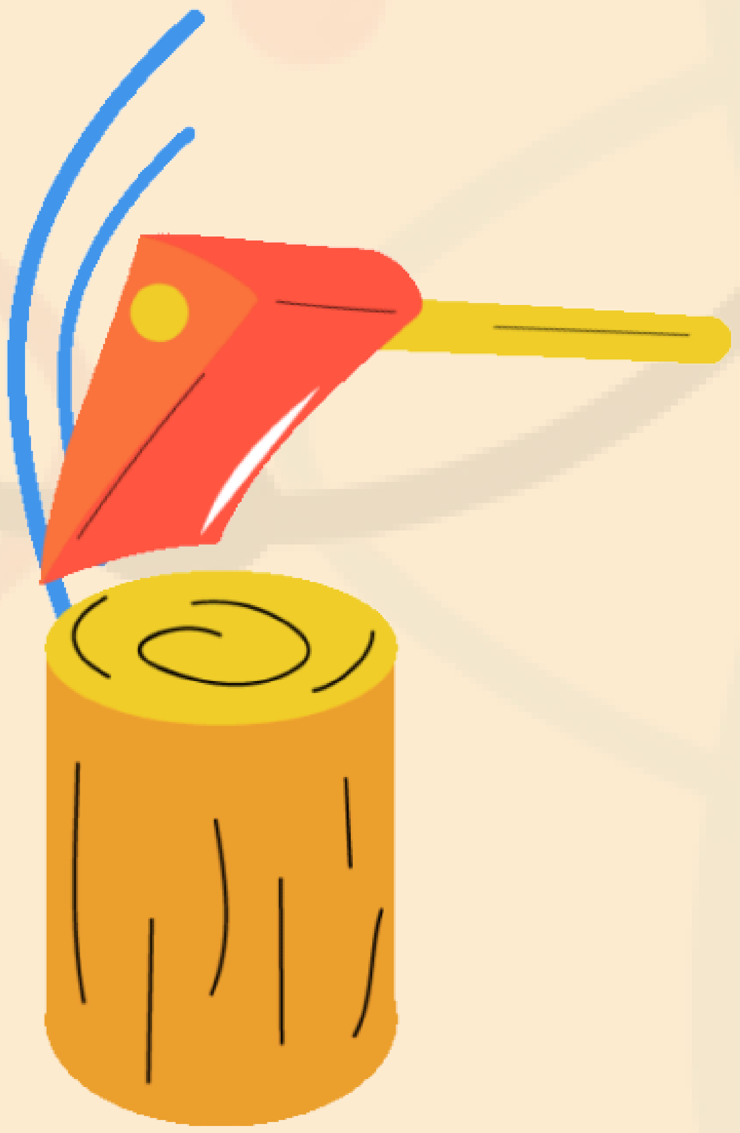


WHAT ARE ATOMS AND MOLECULES?

Atoms are tiny building blocks that make up everything around us. When atoms join together, they form molecules!

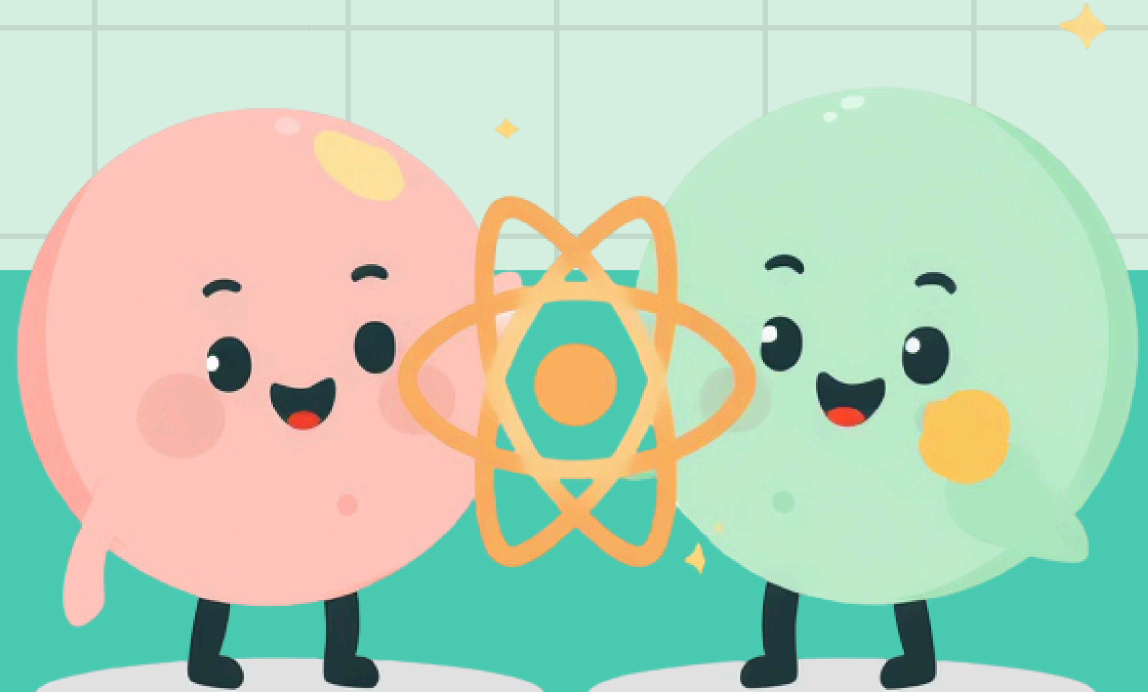
WHY DO ATOMS BOND?

Atoms want to be stable and happy! Their outer electrons (called valence electrons) want to have a full shell. When atoms bond together, they share or transfer electrons to feel complete.



INTRODUCING CHEMICAL BONDS

Chemical bonds are special forces that hold atoms together! There are two main types of bonds we will learn about: Ionic bonds and Covalent bonds. Think of bonds like invisible glue that keeps atoms connected!

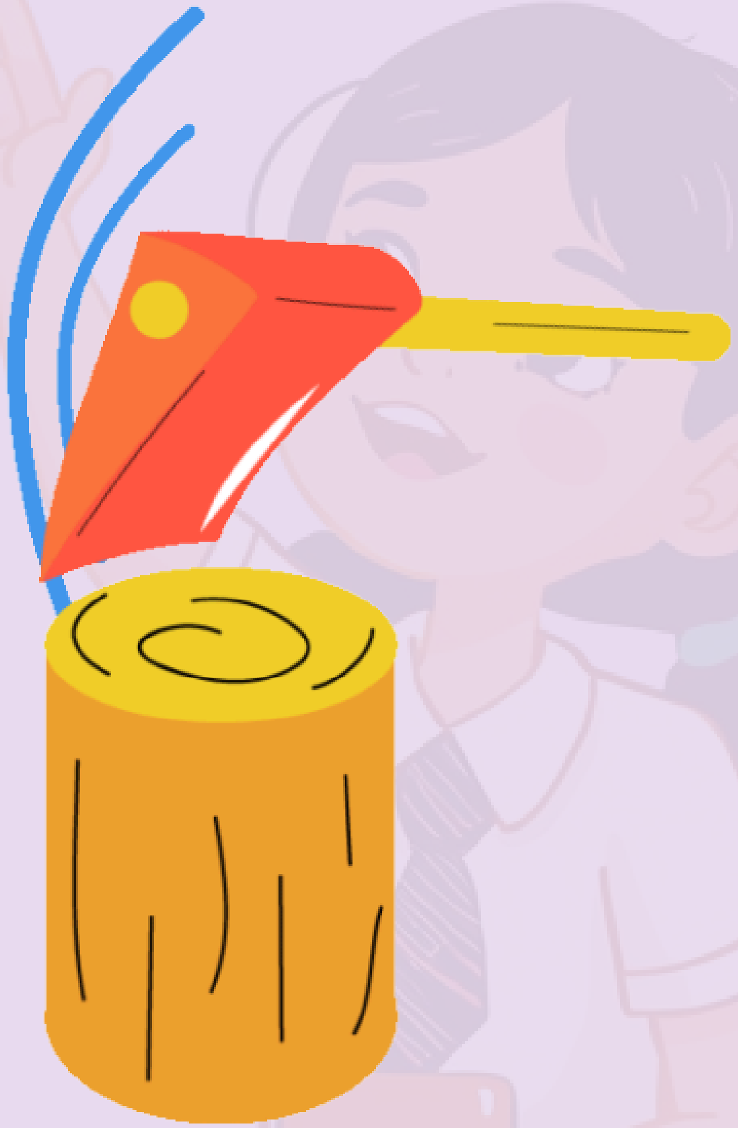


WHAT IS AN IONIC BOND?

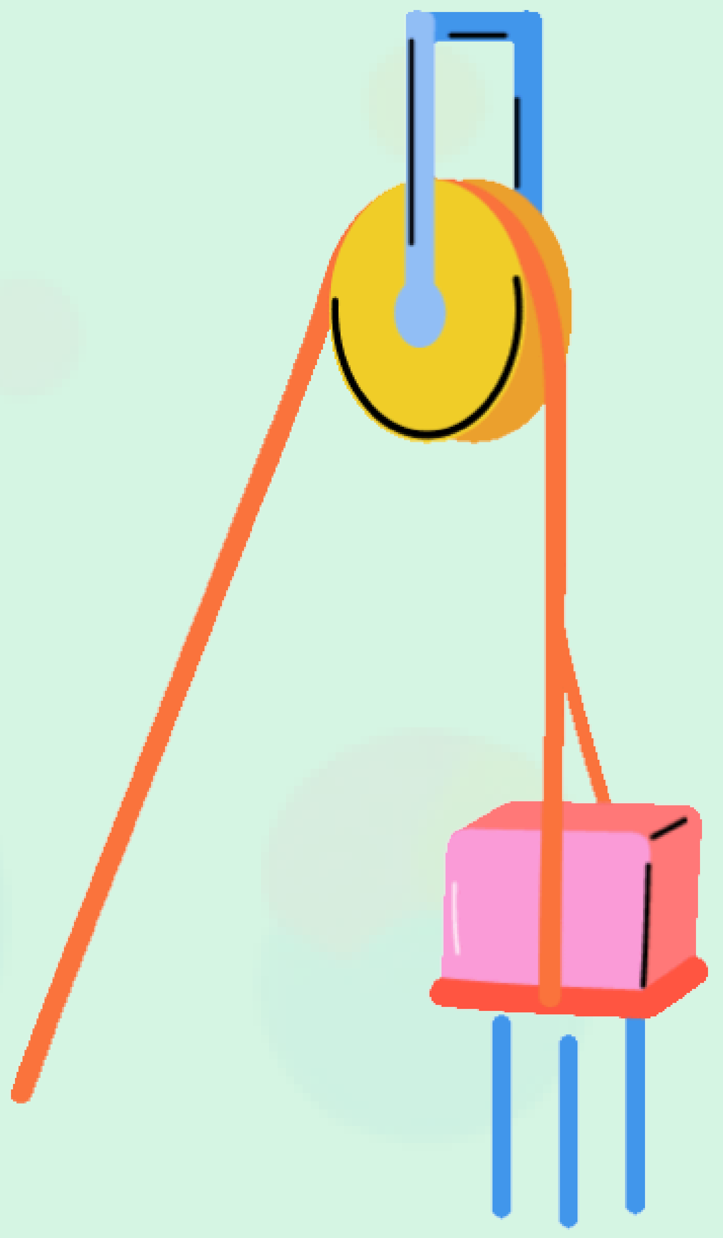
Ionic bonds form when atoms transfer electrons from one to another. One atom becomes positive (+) and the other becomes negative (-). Example: Sodium (Na) gives an electron to Chlorine (Cl) to form table salt (NaCl)!

VISUALIZING IONIC BONDS

Watch how sodium gives its electron to chlorine! Na becomes positive (+) and Cl becomes negative (-). Together they form salt!



Let's share!

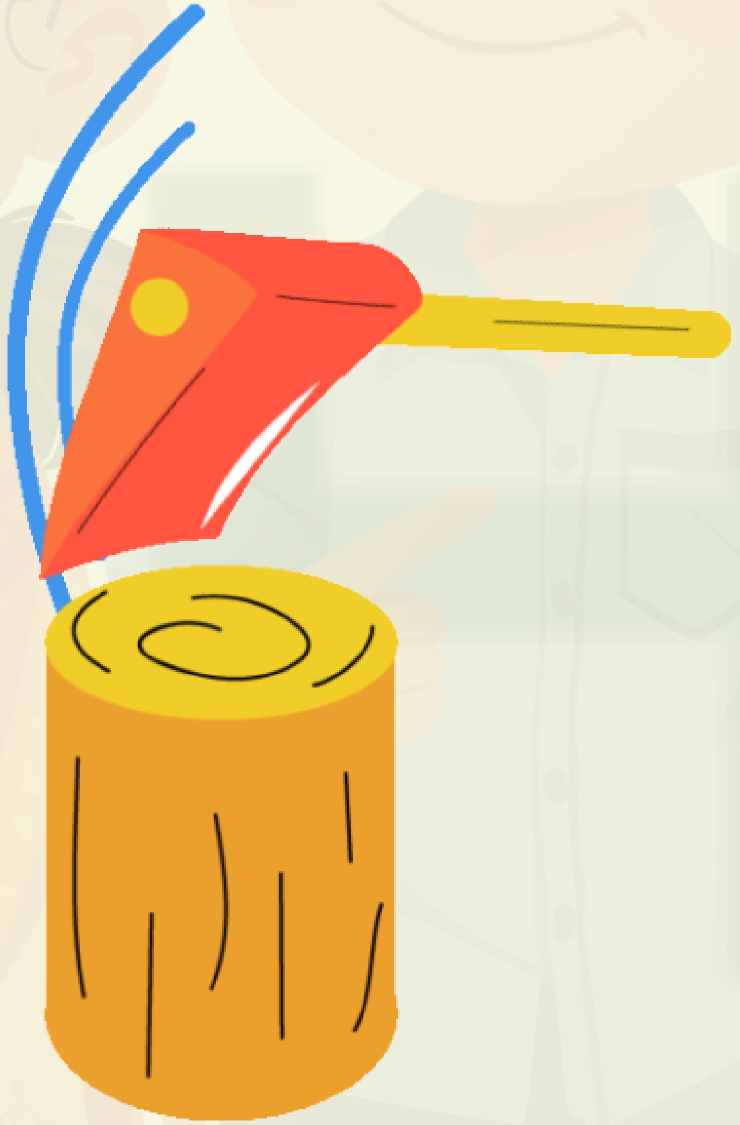


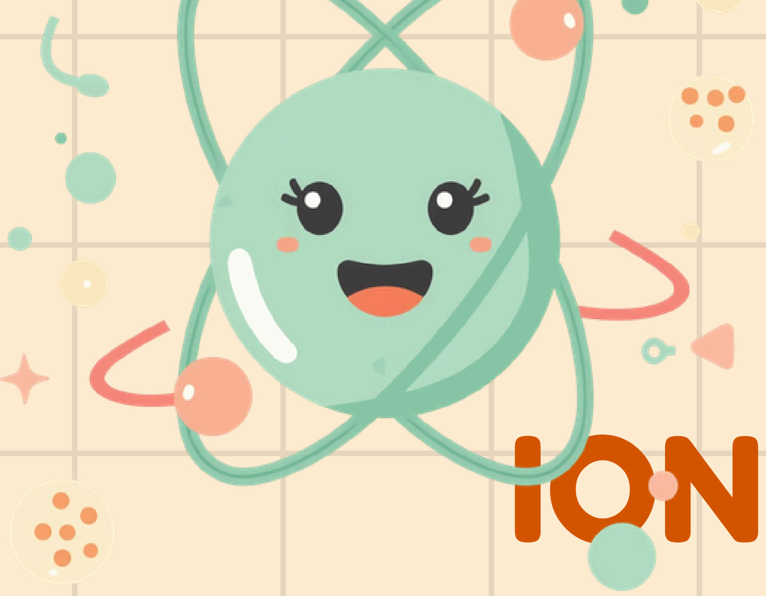
WHAT IS A COVALENT BOND?

Covalent bonds form when atoms share electrons instead of transferring them. Both atoms stay neutral but become stable by sharing!

VISUALIZING COVALENT BONDS

Watch how two hydrogen atoms share their electrons to form a stable H2 molecule. Unlike ionic bonds that transfer electrons, covalent bonds are all about sharing!





IONIC VS COVALENT: KEY DIFFERENCES

1

Ionic Bond: Transfers electrons, forms charged ions, metal + non-metal

2

Covalent Bond: Shares electrons, neutral molecules, non-metals only

3

Key Difference: Ionic bonds are strong crystals, covalent can be single or double bonds

EXAMPLES AROUND US

Table Salt (NaCl) – Ionic bond

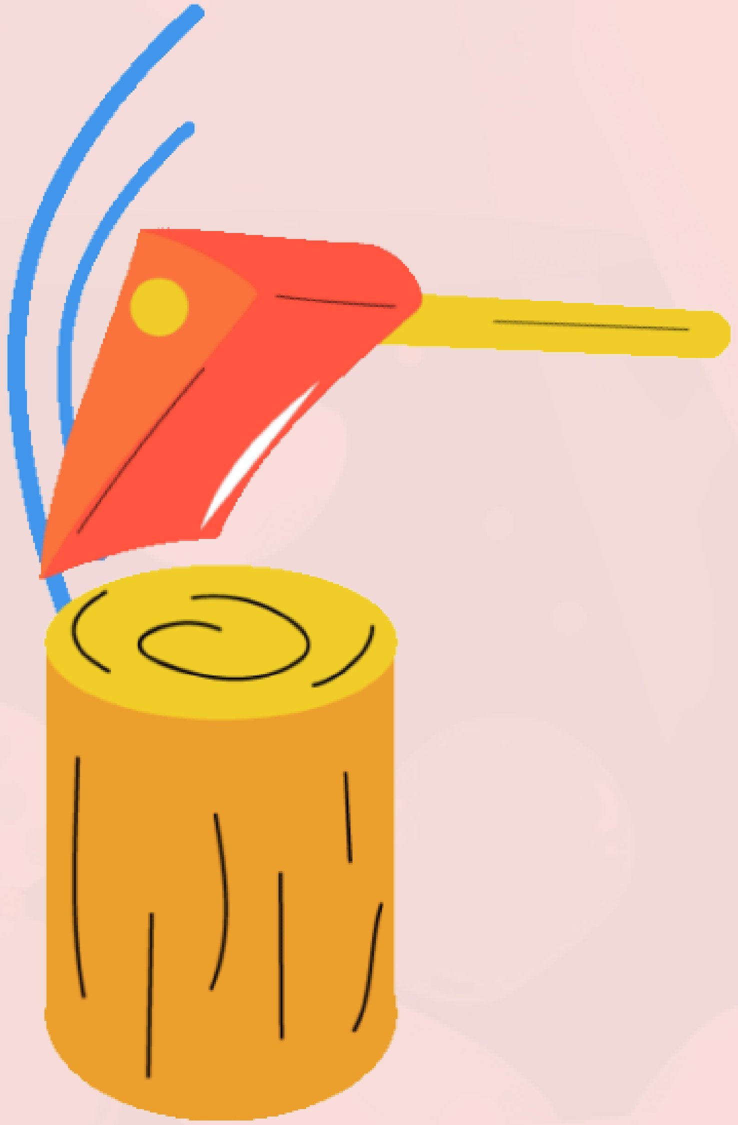
Water (H_2O) – Covalent bond

Sugar ($C_{12}H_{22}O_{11}$) – Covalent bond

Baking Soda ($NaHCO_3$) – Ionic bond

WHY DOES IT MATTER?

Chemical bonds determine important properties like melting point, solubility, and electricity conduction. This explains everyday things like why salt dissolves in water and why water stays liquid at room temperature!



INTERACTIVE QUIZ TIME!

1. Which bond is formed when electrons are shared?

2. What charge does an atom have after losing an electron?

3. Is table salt (NaCl) ionic or covalent?

QUIZ ANSWERS

1. Sharing electrons → Covalent bond

2. Losing electron → Positive charge (cation)

3. Great job! Keep exploring chemistry!

SUMMARY: WHAT WE LEARNED

1

Atoms bond to become stable and happy

2

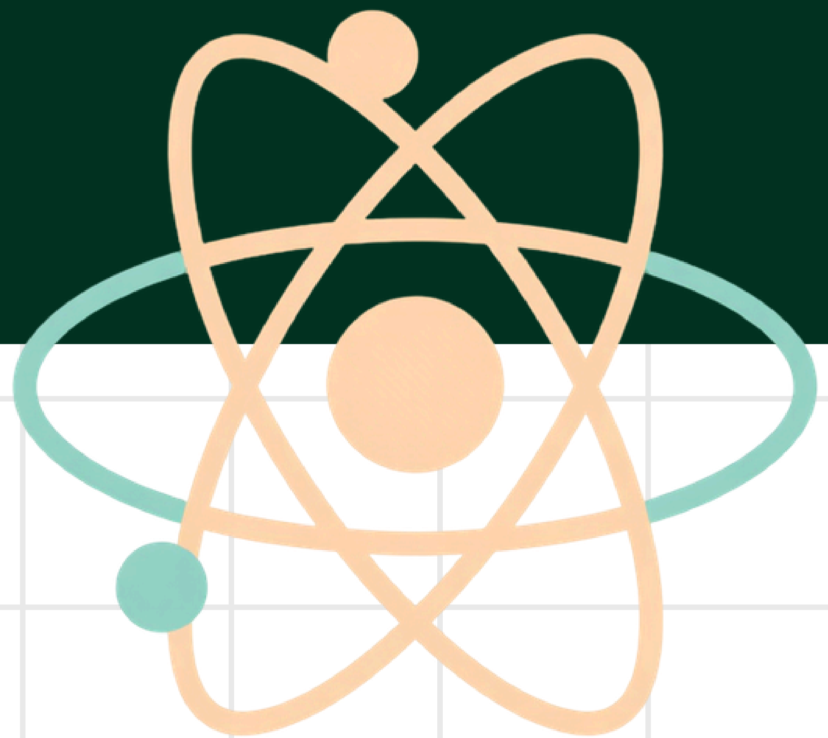
Ionic bonds transfer electrons, covalent bonds share them

3

Different bonds create different properties in everyday life

KEEP EXPLORING CHEMISTRY!

You are now a chemical bonding expert! Try these fun experiments at home: grow salt crystals, make slime to see covalent bonds in action, or dissolve sugar in water. Keep asking questions and exploring the amazing world of chemistry!



THANK YOU & QUESTIONS?

We hope you enjoyed learning about chemical bonding! Feel free to ask any questions.